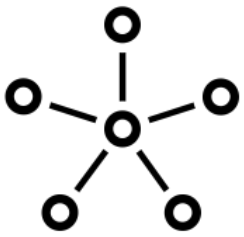
A 2.2.1 Network Topologies



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Network topology describes how devices connect in a network. Common types include star (central hub), mesh (fully interconnected), and hybrid (mix of topologies). Factors like cost, speed, and reliability influence topology choice. Star suits homes, mesh suits critical systems, and hybrids suit large networks. Emerging trends like SDN and cloud computing are shaping future network designs for increased flexibility and scalability.

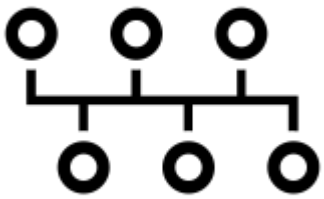
Introduction

Network architecture refers to how computers and other devices are connected and how they communicate. Understanding different network topologies and their characteristics is crucial in designing efficient and reliable networks. This section explores various network topologies' functions and practical applications, focusing on star, mesh, and hybrid configurations. We will also examine the factors influencing network design and how these topologies are applied in various settings.

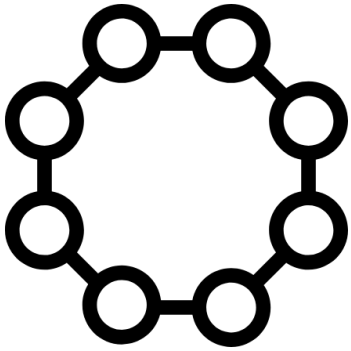
What are the functions and practical applications of different network topologies?

A network topology defines the physical or logical arrangement of nodes (devices) and connections in a network. Different topologies offer unique advantages and disadvantages, making them suitable for specific applications. Here are some common types:

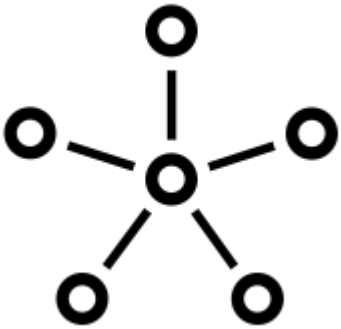
Bus Topology: All devices are connected to a single cable (the bus). It's simple but prone to congestion and failure if the cable breaks.



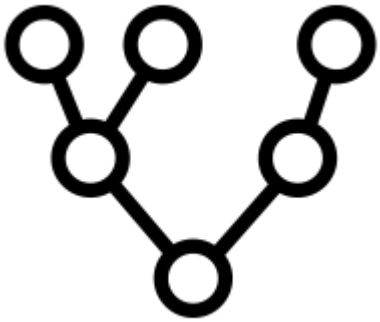
Ring Topology: Devices are connected in a closed loop (ring). Data travels in one direction, reducing collisions. However, a single node failure can disrupt the entire network.



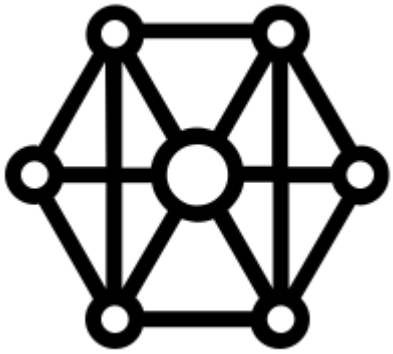
Star Topology: All devices connect to a central hub or switch. This offers better performance and easier troubleshooting, but the hub becomes a single point of failure.



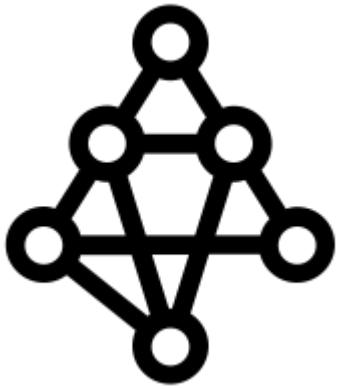
Tree Topology: A hierarchical arrangement where a root node connects to multiple branches, forming a tree-like structure. It's suitable for larger networks but can be complex to manage.



Mesh Topology: Every device connects to every other device, providing high redundancy and fault tolerance. However, it's expensive and complex to implement.



Hybrid Topology: Combines two or more topologies to leverage their strengths. This allows for flexibility and customisation but requires careful planning.



What are the characteristics of star, mesh, and hybrid topologies?

Star Topology:

- Centralised: All devices connect to a central hub.
- Easy to install and manage.
- Single point of failure (the hub).
- Suitable for homes and small offices.

Mesh Topology:

- Decentralised: Every device connects to every other device.
- Highly reliable and fault-tolerant.
- Expensive and complex to implement.
- Suitable for critical applications where downtime is unacceptable.

Hybrid Topology:

- Combines the strengths of different topologies.
- Flexible and adaptable to specific needs.
- Can be more complex to design and manage.

- Commonly used in larger networks.

How do factors like reliability, transmission speed, scalability, data collisions, and cost influence network design?

Reliability: The ability of the network to function consistently without failures. Mesh topologies offer high reliability, while bus topologies are less reliable.
Transmission Speed: The rate at which data is transferred across the network. Star topologies generally offer good speed, while bus topologies can suffer from congestion.
Scalability: The ability to easily expand the network by adding more devices. Star and hybrid topologies are highly scalable, while ring topologies are less so.
Data Collisions: Occur when multiple devices try to send data simultaneously on a shared medium. Bus topologies are prone to collisions, while star topologies minimise them.
Cost: The overall expense of implementing and maintaining the network. Bus topologies are generally the cheapest, while mesh topologies are the most expensive.

How do network topologies apply in different settings, such as home and small office environments where reliability is critical, or larger settings like corporations, government departments, and college campuses?

Home and Small Office: Star topologies are common due to their reliability, simplicity, ease of management, and adequate speed.
Corporations: Hybrid topologies are often used to combine the strengths of different topologies and accommodate various needs within the organisation.
Government Departments: Mesh topologies might be employed for critical systems where high reliability and security are paramount.
College Campuses: A combination of star and tree topologies is common to provide connectivity to different buildings and departments while maintaining manageability.

Key Questions

Can you answer these key questions relating to network topologies? Try to think and write your answers down after reading the text above or researching. Then you can click the eye icon to see some possible answers.

What are the advantages and disadvantages of each network topology?

How does the choice of topology affect network performance and reliability?

What factors should be considered when choosing a network topology for a specific setting?

Can you provide real-world examples of where different topologies are used?

How do network topologies impact data security and privacy?

What are the emerging trends in network architecture, and how might they influence future network design?

Quiz

Take this quiz to test your knowledge of network topologies.

1

Which network topology connects all devices to a single central hub?

- **A.** ○ Mesh
- **B.** ○ Star
- **C.** ○ Bus
- **D.** ○ Ring

2

Which topology is MOST vulnerable to a single point of failure affecting the entire network?

- **A.** ○ Bus
- **B.** ○ Star
- **C.** ○ Mesh
- **D.** ○ Hybrid

3

Which topology offers the HIGHEST level of redundancy and fault tolerance?

- **A.** ○ Ring
- **B.** ○ Star
- **C.** ○ Bus
- **D.** ○ Tree

4

In which topology does data travel in a single direction around a closed loop?

A. ○ Star

B. ○ Tree

C. ○ Bus

D. ○ Ring

5

Which factor is LEAST important when choosing a network topology for a small home network?

A. ○ Redundancy

B. ○ Cost

C. ○ Speed

D. ○ Scalability

6

Which topology is MOST suitable for a large organization with multiple departments and buildings?

A. ○ Hybrid

B. ○ Star

C. ○ Ring

D. ○ Bus

7

What is a disadvantage of a mesh topology?

A. ○ Slow transmission speed

B. ○ High complexity

C. ○ Limited scalability

D. ○ Low reliability

8

Which topology is often used in wireless networks to extend coverage and improve signal strength?

A. ○ Bus

B. ○ Ring

C. ○ Tree

D. ○ Mesh

9

Which emerging trend allows for centralized control and management of network resources through software?

A. ○ Software-Defined Networking (SDN)

B. ○ Internet of Things (IoT)

C. ○ Network Function Virtualization (NFV)

D. ○ Cloud computing

10

Which topology is MOST likely to experience data collisions?

A. ○ Star

B. ○ Tree

C. ○ Mesh

D. ○ Bus